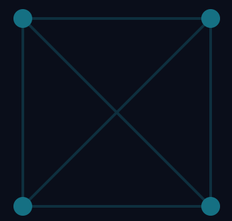
LABLAB.AI · AGENTIC ECONOMY ON ARC · APR 2026

Signal Mesh on Arc

Specialist trading agents author sub-cent USDC payments — Circle developer-controlled wallets execute on Arc. USDC is the gas, the payment, and the unit of account.



Leewonwuk · skyskywin@gmail.com · github.com/Leewonwuk/signal-mesh-arc

USDC is the gas.
On Arc.

The hidden human in every "agent economy".

Same story, two different currencies on the ledger — until Arc.

OTHER CHAINS

USDC earned — agent does work

↓

ETH / SOL / MATIC paid in gas

↓

A human has to keep refilling the gas wallet

This is not an agent economy. It's a human economy with an LLM on top.

ARC

USDC earned — agent does work

↓

USDC paid in gas

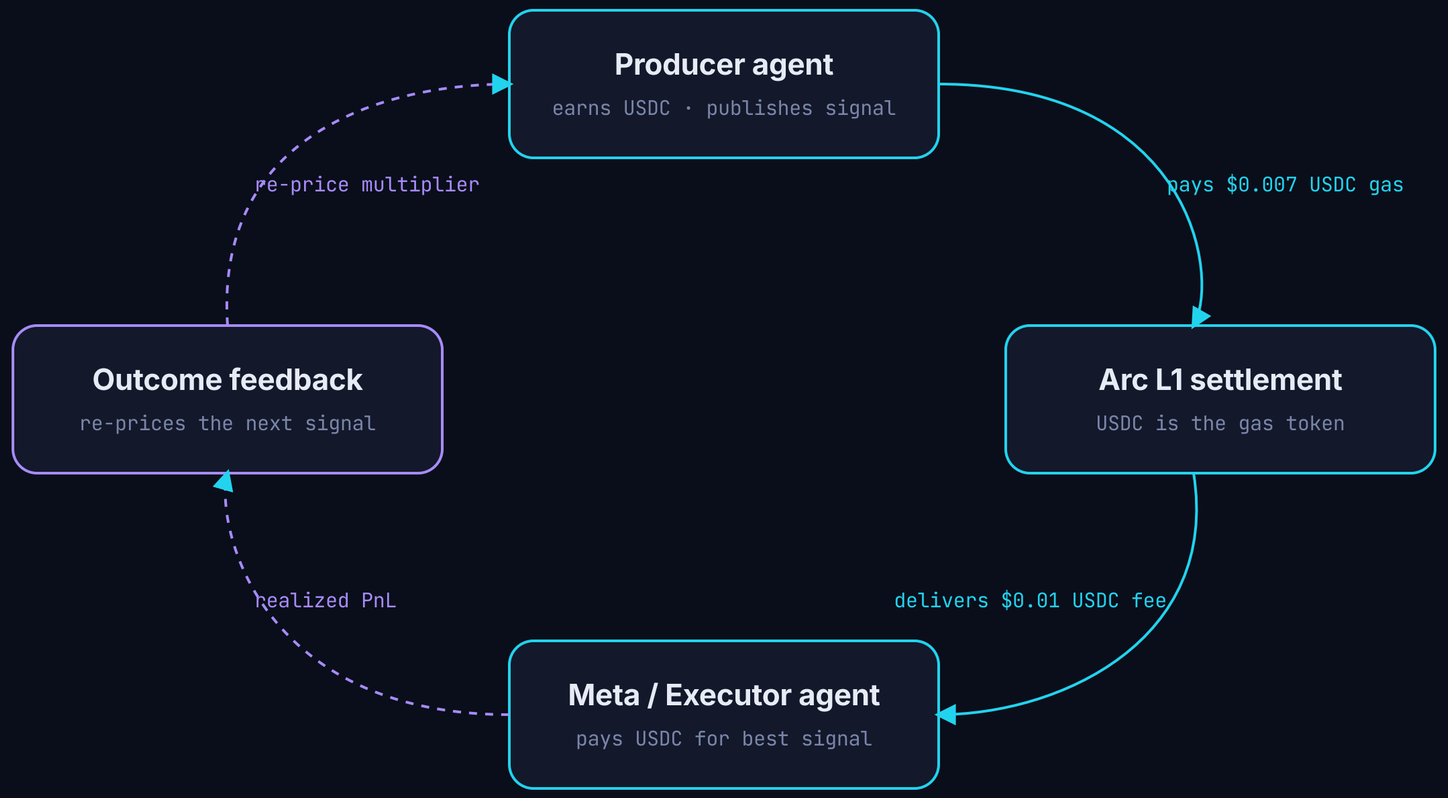
↓

USDC received by the next agent

Loop closes. No human in the middle.

Arc makes USDC the gas token.

Same currency in, same currency out. The loop closes.



Gas and revenue in the same unit → no ETH treasury, no gas-refill loop, no per-tx top-up.

Four agents. One bridge. Arc testnet.

Everything below is wired and sending real USDC on-chain.

PRODUCERS · 3

- **Kimchi**
  KRW premium · synthetic AR(1)
- **Dual-Quote**
  Live v1.3 production bot · 1s tape
- **Funding**
  Binance fapi · 1,355 real 8h cycles

BRIDGE

Signal registry

- **Pricing** — confidence × notional × take-rate
- **Routing** — meta-policy picks winner
- **Outcome feedback** — realized PnL back to producer

`bridge/src/index.ts · TypeScript`

CONSUMERS · 2

- **Meta agent**
  Gemini · arbitrates signals
- **Executor**
  Paper trade + settle USDC on Arc
- **Arc L1 testnet**
  `60+ real tx · chainId 5042002`

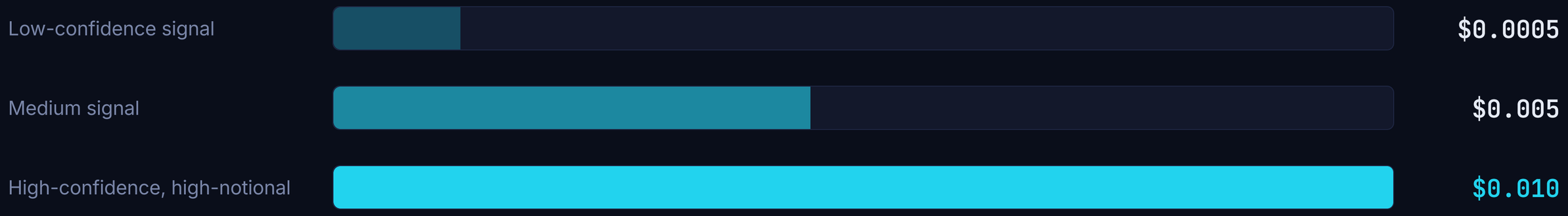
Producers compete. Each runs a different strategy. Every signal has a price attached.

Meta arbitrates. LLM picks the winner, weighted by which producer has been right lately.

Executor pays. Real USDC settles on Arc. Price scales with signal quality.

The USDC amount on-chain IS the quote.

Not a receipt constant. A live signal-quality score.




```
price = confidence × notional × |premium| × take_rate  →  clipped to [$0.0005, $0.010]
```

A producer that keeps being right gets paid more. A producer that keeps being wrong gets paid less. The Q-table sets the multiplier.

THE QUESTION BEHIND THE BACKTEST

Which strategy should trade right now?

V1

Kimchi Premium

KRW ↔ USDT cross-exchange spread. Wakes up when Korea gets excited.

V2 · CURRENTLY LIVE

Dual-Quote Arb

USDT ↔ USDC intra-Binance spread. Fast, thin, high-frequency.

LIVE ON EC2 PRODUCTION

V3

Funding-Rate Basis

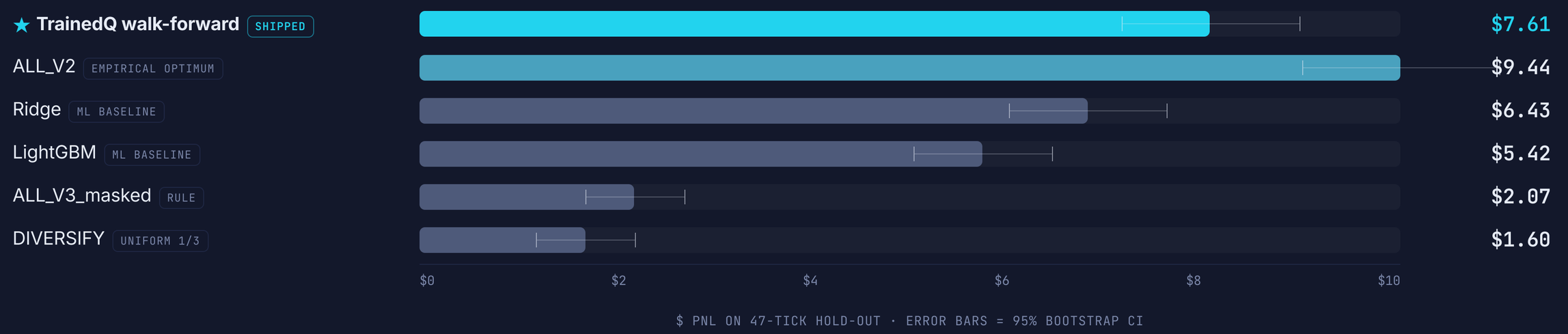
Perp funding cadence arb. Wakes up every 8 hours.

We don't pick. We learn.

An agent watches market conditions, scores outcomes, and picks the strategy the data says works right now.

The learned policy matches the empirical best.

47-tick out-of-sample hold-out · split `cal:178` / `train:46` / `test:47` · calibration frozen before training opens.



- ✓ Statistically ties the empirical optimum `p = 0.49 vs ALL_V2`
- ✓ Beats diversify `p = 0.012 vs DIVERSIFY`

ROUND 2. ROUND 1 FAILED — REWARD FUNCTION WAS BROKEN. FIXED LIVE, DOCUMENTED IN SUBMISSION.MD §11.B.1.

Nothing is hidden. Here is exactly what is real.

Every payload on the dashboard carries its provenance tag. Judges should be able to fact-check.

V1 · KIMCHI

SYNTHETIC

AR(1) · SEED COMMITTED

AR(1) mean-reverting premium series with committed seed.

ρ = 0.62 · σ = 0.0031 · seed=2026_04_18

Real cross-venue Upbit KRW + Binance USDT data wasn't available in time. Clearly labeled in every payload and on the dashboard.

V2 · DUAL-QUOTE

REAL + BOOTSTRAP

1-DAY TAPE · RESAMPLED

1 day of live v1.3 production 1-second tape. Bootstrap-resampled within-regime to fill a 90-day window.

2026-04-19 · 86,400 bars · resample within-regime

The 1-second bars are prices the production bot actually saw. Bootstrap is documented as bootstrap, not fabricated.

V3 · FUNDING

REAL

PUBLIC REST · REPRODUCIBLE

Binance

fapi/v1/fundingRate

public REST · 2026-01-21 → 2026-04-21.

5 symbols × 271 ticks = 1,355 real 8h cycles

No API key required. Anyone can reproduce from scratch.

Backtest convergence reproducible from public data alone.

What we used. What we declare.

Circle product inventory and custody model — stated plainly.

Arc testnet

60+ settlement tx. `chainId 5042002`. USDC as the gas token.

USDC

Unit of account for fees, payments and gas. Dual decimal (18 native / 6 ERC-20) handled.

Circle Wallets (SCA)

Developer-Controlled Wallets. 4 agent wallets via Wallet Sets. Signing via `entitySecretCiphertext`.

Developer API transfers

`/v1/w3s/developer/transactions/transfer` posts the 60-tx burst. EIP-3009 path validated but mempool-deferred.

Trust model, declared.

4 agent wallets are Circle-custodied SCAs, signed by `entitySecretCiphertext`. Three concrete failure modes:
(a) Entity-secret leak → all 4 wallets compromised simultaneously (no per-wallet signing isolation).
(b) Circle freezes any wallet (KYC / sanctions / ToS) → outcome loop halts, reliability scoring breaks.
(c) API rate-limit at sub-cent cadence = de-facto censorship, no protocol-level recourse.
Mainnet roadmap (SUBMISSION §12): Circle Wallets MPC + raw EIP-3009 on normalized mempool.

What's new here.

Three things we haven't seen in other agent-economy demos.

01

Variable on-chain price

The USDC amount you see transferred is the price the market set for that specific signal — not a receipt constant.

see ArcScan: any two tx of ours have different USDC amounts

02

Closed outcome loop

Realized PnL retroactively re-prices future signals from the same producer. Variable pricing runs live in the executor's paper path; the 60-tx Circle burst is the evidence dump on the same Arc rails — fixed cadence, real hashes.

bridge/src/index.ts · /signals/outcome → /producer/reliability (N=200)

03

Meta-policy above the market

A learned allocator picks which of three strategies even gets to trade. We use learning to validate the rule, not to replace it.

9 × 7 state-action table · walk-forward tested

Most "agent economy" demos stop at "LLM + wallet." We didn't.

Zero humans in the loop.

Open the repo. Fire the demo. Watch the allocator pick.

SIGNAL MESH ON ARC

GITHUB
github.com/Leewonwuk/signal-mesh-arc

PITCH VIDEO
YouTube · unlisted

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